expect Module

اگر زمانی خواهیم که یک command را روی یک host انجام بدیم که نیازمند یک پاسخ از جانب ماست از expect استفاده می کنیم

- The ***expect*** module executes a command and responds to prompts.
- shell variables or operations such as <, >, |, and & will not work as it will not be processed by the shell.
- Requirements:
  - Python >=2.6
  - Pexpect >= 3.3
- On the hosts:

Yum install epel-release
Yum install python-pip
Pip install pexpect
Or
apt-get install python3-pexpect

```
- name: expect module
  expect:
    command: passwd LPI
    responses:
      password: "Anisa"
    no_log: true
```

The command to run

Mapping of expected string to respond with

جلوگیری از log گرفتن هنگام Run

Script Module

مایک script نوشتیم که روی خود سرور Ansible هست و بصورت Remote این رو اجرا کنیم

- Copy a script remotely to a machine and execute it. This module does not require python on the remote system

```
- name: script module
  script: test.sh
  args:
    creates: /home/TestDir3
    removes: /home/TestDir2
  register: scp_var
  tags: [script_mod]

- debug: msg={{ scp_var.stdout_lines }}
  tags: [script_mod]
```

When it already exists, the command will not be run (creates)

اگر این آدرس وجود داشت اجرا نکن

When it does not exist, the command will not be run (removes)

اگر این آدرس وجود نداشت این اسکریپ اجرا شود

لاگ رو داخل این var ریجستر کن و توخروجی نشان بده


```
- name: "Run Scripts"
  script: /home/ansible/provision/roles/Myprj/files/script.sh
  removes: /home/dir1
  tags: sh
```

Shell Module

مثل ماژول Command هست ولی محدودیت‌های آن را ندارد

- uses a shell (/bin/sh) to run the commands. Shell environment variables and features can be used.

```
- name: shell module
  shell: echo "Test Shell module!" > TestFile
  args:
   chdir: /home/TestDir
   creates: /home/TestDir3
   removes: /home/TestDir2
  tags: [shell_mod]
```


Exercise 10: shell module

1. Use ansible ***shell*** module to run the following commands
   1. Cat /home/dir1/file1
   2. Cat < /home/dir1/file1
   3. df -h
   4. ps aux | grep ansible
   5. Curl -v anisa.co.ir

```
- name: "Shell"
  ansible.builtin.shell: >
    echo "Hello" > /home/dir1/file1 && cat /home/dir1/file1 && df -h && ps aux |
grep -i ansible && curl -v anisa.co.ir
  register: logss
  tags: she

- name: "ppyrt"
  debug:
    msg: "{{logss}}"
  tags: p
```

2. Print the output of the above commands and check it

Raw Module

- Executes a command in a few cases:
- Run commands on systems which don't have any python installed.
- Install python-simplejson on older (Python 2.4 and before) hosts that need it as a dependency to run modules.
- Speak to any devices such as routers that do not have any Python installed.

```
- name: raw mrdule
  raw: echo "test raw module" > /home/test
  tags: [raw_mod]
```


Copy Module

- Copy a file from the local or remote machine to a location on the remote machine.

```
- name: copy module
  copy:
    src: myfile
    dest: /home/TestDir2
    owner: root
    group: root
    mode: 0644
    backup: yes
    force: yes
    tags: [copy_mod]
```

Copy.Backup

If yes, create a backup file including the timestamp information

Copy.force

If yes, replace the remote file when it is different than the source. If no, it will be transferred only the destination dose not exist.

Copy.remote_src

Influence whether src needs to be transferred or already is present remotely.

Exercise 11: copy module

1. Put a file named myfile in the files directory of your playbook
2. Copy the file to /home on the both remote machines
3. Set backup to yes and try to change the file content and run the playbook again
4. Try to copy /home/myfile to /opt on remote machines (src and dest are both on the remote machines)
5. Set the force to no and check the result



Fetch Module

زمانی که بخواهیم یک فایل از host به سرور اصلی ما منتقل شود برعکس cp

- Just like copy module, but in reverse. Fetch files from remote machines and store them locally in a file tree, organized by hostname.

```
- fetch:
    src: /tmp/somefile
    dest: /tmp/fetched
    validate_checksum: yes
```

اگه 2 تا file همسان باشند نمیره

Validate_Checksum

If yes, Verify that the source and destination checksums match

fail_on_missing

The task will fail if the source file (eg. log files) is missing.

Flat

override the default behavior of appending hostname/path/to/file to the destination.



Group Module

- Manage groups on a host

```
- name: group module
  group:
    name: DevOps
    state: present
  tags: [group_mod]
```



User Module

- Manage users on a host

```
- name: user module
  user:
    name: Sayahi
    group: DevOps
    state: present
    shell: /bin/bash
    password: 123
  tags: [user_mod]
```



Exercise 12: user and group module

1. Add a group named anisa on both remote machines
2. Add a user named devops to the anisa group
    1. Set the shell to /bin/bash
    2. Set the home to /tmp/devops
3. Change the user password with expect module



Zypper_repository Module

- Add or remove Zypper repositories

```
- name: --- zipper_repository module ---
  zypper_repository:
    name: ansible
    repo: /opt/Ansible
    state: present
  tags: [zypp_mod]
```

توضیحاتی که SUSE یا openSUSE توانیم یک repo حذف یا اضافه کنیم



Zypper Module

- Manage packages on SUSE and openSUSE using the zypper and rpm tools.

```
- zypper:
  name:'http://nginx.org/packages/sles/12/x86_64/RPMS/
nginx-1.8.0-1.sles12.ngx.x86_64.rpm'
  state: present
```

```
- name: Install Ansible --- python-SetupTools
  zypper:
  name: {{ Ansible_Repo.Repo_Dir }}/python-
setuptools-18.0.1-1.2.noarch.rpm
  state: latest
  disable_gpg_check: yes
```



yum_repository Module

ماژول ماو repo با Update

- Add or remove YUM repositories in RPM-based Linux distributions

```
- name: --- yum_repository module ---
  yum_repository:
   name: epel
   description: extra packages
   baseurl: https://download.fedoraproject.org/pub/epel/$releasever/
$basearch/
   gpgcheck: no
   enabled: yes
   state: present
  tags: [yum_repo_mod]
```



yum Module

- Only update installed packages. (Don't install packages)

```
- name: --- yum module (update Nginx) ---
  yum:
    name: nginx
    state: latest
    update_only: yes
  tags: [yum_mod]
```

- Exclude packages from upgrading

```
- name: --- yum module (upgrade all packages except kernel packages) ---
  yum:
    name: '*'
    state: latest
    exclude: kernel*
```



yum Module

- List available Ansible packages

```
- name: --- list available Ansible packages ---
  yum:
    list: ansible
  register: yum_out
  tags: [yum_list]

- debug:
    msg: '{{ yum_out }}'
  tags: [yum_list]
```



Exercise 13: yum module - Install Apache

1. List available ***httpd*** packages on both remote machines
2. **Install** apache on the hosts
3. Run the playbook again and only **update** the apache
4. Set *state: latest* and ***security****: yes* so only installs updates that have been marked security related
5. Set *state: latest* and ***bugfix****: yes* so only installs updates that have been marked bug fix related



apt Module

- Manages apt packages (Install, upgrade, downgrade, remove, and list packages on Debian/Ubuntu)

```
- name: --- Install nginx on ubuntu ---
  apt:
   name: nginx
   state: present
  tags: [apt_mod]

- name: --- Install a .deb package —
  apt:
   deb: /opt/package.deb
  tags: [apt_mod]
```



apt Module

```
- name: --- Remove nginx on ubuntu ---
  apt:
   name: [nginx,nginx-common]
   state: absent
   purge: yes
  tags: [apt_mod]

- name: --- remove unused dependency packages —
  apt:
   autoremove: yes
  tags: [apt_mod]
```

apt.purge

Purge configuration files if the module state is set to *absent*.

apt.autoremove

remove unused dependency packages



Template Module

2 روش برای تغییر config file هادر host داریم ← فایل رو ادیت کنیم / فایل جدید رو جایگذاری کنیم

- Templates are processed by the **Jinja2** templating language.

```
- name: Template Module
  template:
   src: nginx.conf.j2
   dest: /etc/nginx/nginx.conf
   owner: root
   group: root
   backup: yes
   force: yes
  tags: [template_mod]
```

کانفیگ فایل ← src
کجا فایل قرار بگیرد ← dest

Template.Backup

If yes, create a backup file including the timestamp information

Template.force

If yes, replace the remote file when it is different than the source. **If no**, it will be transferred only the destination dose not exist.



Exercise 14: template module - configure Apache

1. Place the httpd.conf in the template directory
2. Change the listen port to 8080
3. Try configure the apaches on the remote machines with the new httpd.conf
4. Set backup: yes and change the httpd.conf.j2 and run the playbook again
5. Set force: no. change the httpd.conf.j2 and run the playbook again



Lineinfile Module

- search a file for a line

```
- name: lineinfile module
  lineinfile:
    path: /etc/nginx/ngix.conf
    insertafter: '      listen      80 default_server;'
    line: '      listen      8080 default_server;'
```

Lineinfile.insertbefore

the line will be inserted before the last match of specified regular expression.

Lineinfile.insertafter

the line will be inserted after the last match of specified regular expression.

Lineinfile.line

The line to insert/replace into the file.



Exercise 15: lineinfile module - configure Apache port

1. Use lineinfile module to edit httpd.conf
    1. Set insertafter: '^#Listen '
    2. Set line: 'Listen 8080'
    3. Set backup: yes
2. Run the playbook and check the configuration file on the hosts
3. Try to remove the line you've just added (state: absent)



Replace Module

- replace all instances of a pattern within a file

```
- name: Replace module
  replace:
    path: /etc/nginx/nginx.conf
    regexp: '      listen      80 default_server;'
    replace: '      listen      8080 default_server;'
    after: 'Start after line.*'
    before: 'Start before line.*'
    backup: yes
```

Replace.regexp

The regular expression to look for in the contents of the file

Replace.replace

The string to replace regexp matches

Replace.Backup

Create a backup file including the timestamp information



Exercise 16: replace module - change Apache port

1. Use replace module to edit httpd.conf
    1. Set regexp: 'Listen 80'
    2. Set replace: 'Listen 8080'
    3. Set backup: yes
2. Run the playbook and check the configuration file on the hosts



Service Module

برای مدیریت سرویس start - stop ...

- Controls services on remote hosts.

```
- name: Enable Nginx Service
  service:
    name: nginx
    enabled: yes
  tags: [enable_nginx]

- name: Start Nginx service
  service:
    name: nginx
    state: started
  tags: [start_nginx]
```

Service.enabled

Whether the service should start on boot

Service.state

started, stopped, restarted, reloaded

Service.sleep

If the service is being restarted then sleep this many seconds between the stop and start command.



Exercise 17: service module - enable and start Apache

1. Enable and start apache service
2. Check the apache in a browser
3. Try to change the listen port in the httpd.conf
4. restart the apache service
5. Check it again in your browser



مجموعه‌ای از فایل‌ها را آرشیو کند
فشرده سازی

Archive Module

- Packs an archive. It will not copy the source file from the local system to the target before archiving. Requires tarfile, zipfile, gzip and bzip2 packages on target host.
- can produce gzip, bzip2, lzma and zip compressed files.

```
- name: archive module
  archive:
    path: /home/TestDir2/*
    dest: /tmp/ArcTestDir2.tar.gz
    remove: yes
    exclude_path:
     - /home/TestDir2/TestFile2
  tags: [archive_mod]
```

Archive.remove

Remove the source file after adding to archive

Archive.exclude_path

Specify the files to exclude from the archive

Archive.format

The type of compression: gz, bz2, zip, tar, xz



Unarchive Module

- Unpacks an archive. By default it will copy the source file from the local system to the target before unpacking.

```
- name: unarchive module
  unarchive:
    src: ArcTest.tar.gz
    dest: /home/TestDir2
    exclude:
     - f2
    keep_newer: no
  tags: [unarchive_mod]
```

Unarchive.remote_src

If yes, indicates the archived file is already on the remote system

Unarchive.keep_newer

Do not replace existing files that are newer than files from the archive

Unarchive.creates

When it already exists, the command will not be run

